TECHNICAL SUPPLEMENT 01 · ENGLISH

Descriptive Statistics of the Evaluated Campaign

Agri-Social Navigation in Vineyard Corridors

Reinaldo Betancourt
Ingrid Nicole Vásconez
Viviana Moya
William Chamorro
Sandra Cano
Marco Antonio Molina
Juan Pablo Vásconez

Corresponding author: Juan Pablo Vásconez

Study identity	Specification
Campaign / Campaña	EVAL_20261019_112257
Navigation / Navegación	9 × 4 × 10 = 360; SUCCESS 266; TIMEOUT 94
Benchmark	180; 20 per method / por método
Version / Versión	20 October / octubre 2026

Numerical companion to the revised Sensors manuscript. This report uses the identified endpoint dataset and separates it from the older public campaigns. All seven authors are listed in the original manuscript order.

1. Design and estimands

Each of the 36 method-scenario cells has ten attempts. Failed missions remain in the accounting. Completion duration is conditional on SUCCESS; path length and exposure describe the observed attempt window, including partial timeout paths.

Mean = sum(x)/n; sample SD = sqrt(sum((x-mean)^2)/(n-1)). Missing completion duration for an unsuccessful mission is not replaced by 180 s.

Pooled across E1-E4

Method	Success / n	T success [s]	L observed [m]	d min [m]	E1.2 [s]
M0	40/40	95.955 ± 0.049	94.071 ± 0.005	0.796 ± 0.219	26.180 ± 8.510
M1	0/40	--	32.574 ± 5.293	1.083 ± 0.006	147.669 ± 5.205
M2	0/40	--	31.729 ± 4.815	1.089 ± 0.014	147.457 ± 5.263
M3	36/40	125.030 ± 18.688	96.944 ± 1.195	1.128 ± 0.011	5.609 ± 3.237
M4	40/40	117.875 ± 12.997	95.611 ± 0.623	1.116 ± 0.004	8.823 ± 2.945
B1	35/40	115.557 ± 16.489	93.266 ± 7.346	1.095 ± 0.051	9.088 ± 11.386
B2	39/40	101.415 ± 2.860	95.690 ± 1.942	1.023 ± 0.090	18.008 ± 16.551
B3	40/40	102.281 ± 6.155	95.716 ± 0.957	1.091 ± 0.023	5.463 ± 2.262
B4	36/40	105.458 ± 8.370	98.260 ± 11.676	1.281 ± 0.206	0.815 ± 2.229

E1

Method	Success / n	T success [s]	L observed [m]	d min [m]	E1.2 [s]
M0	10/10	95.969 ± 0.046	94.071 ± 0.005	0.745 ± 0.223	35.230 ± 6.285
M1	0/10	--	24.593 ± 1.764	1.080 ± 0.005	155.611 ± 1.606
M2	0/10	--	24.230 ± 1.398	1.079 ± 0.006	155.348 ± 1.875
M3	10/10	121.999 ± 6.046	97.311 ± 0.577	1.123 ± 0.011	6.043 ± 1.472
M4	10/10	110.561 ± 2.684	95.222 ± 0.323	1.115 ± 0.003	7.152 ± 0.938
B1	10/10	106.960 ± 3.066	94.551 ± 0.634	1.113 ± 0.045	4.120 ± 1.734
B2	10/10	99.389 ± 0.320	95.347 ± 0.092	1.035 ± 0.060	7.420 ± 2.863
B3	10/10	99.591 ± 1.522	94.989 ± 0.270	1.100 ± 0.005	4.952 ± 1.866
B4	10/10	100.170 ± 0.778	99.325 ± 0.652	1.471 ± 0.211	0.000 ± 0.000

E2

Method	Success / n	T success [s]	L observed [m]	d min [m]	E1.2 [s]
M0	10/10	95.931 ± 0.046	94.069 ± 0.006	1.077 ± 0.001	18.059 ± 0.071
M1	0/10	--	31.486 ± 0.050	1.087 ± 0.006	148.562 ± 0.051
M2	0/10	--	31.478 ± 0.047	1.086 ± 0.005	148.569 ± 0.047
M3	10/10	113.400 ± 0.945	96.569 ± 0.107	1.135 ± 0.004	3.586 ± 0.183
M4	10/10	110.671 ± 0.426	95.624 ± 0.161	1.116 ± 0.001	8.214 ± 0.229
B1	10/10	112.210 ± 5.036	96.084 ± 0.757	1.106 ± 0.004	7.140 ± 3.753
B2	10/10	98.859 ± 0.071	94.759 ± 0.035	1.074 ± 0.001	15.191 ± 0.129

Method	Success / n	T success [s]	L observed [m]	d min [m]	E1.2 [s]
B3	10/10	99.700 ± 0.212	95.079 ± 0.030	1.103 ± 0.002	4.800 ± 0.153
B4	10/10	104.800 ± 1.714	101.936 ± 0.869	1.153 ± 0.007	0.900 ± 0.125

E3

Method	Success / n	T success [s]	L observed [m]	d min [m]	E1.2 [s]
M0	10/10	95.970 ± 0.046	94.072 ± 0.003	0.639 ± 0.008	32.589 ± 0.567
M1	0/10	--	37.283 ± 0.045	1.077 ± 0.003	143.342 ± 0.051
M2	0/10	--	35.081 ± 0.052	1.109 ± 0.008	143.501 ± 0.202
M3	10/10	116.080 ± 3.592	95.768 ± 0.381	1.131 ± 0.003	4.528 ± 0.879
M4	10/10	110.460 ± 0.549	95.143 ± 0.114	1.118 ± 0.002	6.820 ± 0.567
B1	10/10	110.990 ± 6.819	95.573 ± 0.741	1.107 ± 0.024	6.280 ± 7.462
B2	10/10	102.240 ± 0.578	95.800 ± 0.232	1.040 ± 0.053	15.521 ± 4.721
B3	10/10	99.972 ± 0.047	95.704 ± 0.011	1.085 ± 0.000	4.829 ± 0.065
B4	10/10	101.120 ± 0.103	99.175 ± 0.111	1.158 ± 0.005	0.330 ± 0.048

E4

Method	Success / n	T success [s]	L observed [m]	d min [m]	E1.2 [s]
M0	10/10	95.950 ± 0.054	94.071 ± 0.007	0.724 ± 0.187	18.840 ± 2.064
M1	0/10	--	36.934 ± 0.641	1.086 ± 0.003	143.162 ± 0.679
M2	0/10	--	36.128 ± 1.408	1.082 ± 0.007	142.409 ± 0.470
M3	6/10	164.382 ± 6.828	98.128 ± 1.518	1.124 ± 0.018	8.277 ± 5.336
M4	10/10	139.809 ± 3.399	96.455 ± 0.577	1.113 ± 0.008	13.109 ± 2.820
B1	5/10	148.580 ± 22.366	86.854 ± 13.042	1.055 ± 0.079	18.810 ± 18.536
B2	9/10	105.588 ± 2.151	96.854 ± 3.696	0.943 ± 0.136	33.899 ± 27.124
B3	10/10	109.860 ± 8.753	97.092 ± 0.869	1.074 ± 0.040	7.270 ± 3.716
B4	6/10	122.600 ± 6.078	92.604 ± 23.168	1.342 ± 0.244	2.030 ± 4.343

Interpretation

M0, M4 and B3 complete all 40 assignments each. M1 and M2 complete none under these presets. A smaller partial path is not a shorter completed mission. Pooled conditional means do not remove scenario effects or selection by completion.

Source and scope

Source: data/; results/expected/. PDFs provide the reading reports; CSV/JSON retain numerical inputs and provenance. Scope: evaluated simulation, without new experiments, independent raw-stream reconstruction or physical validation.